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HyperNetwork with Transformer-Based Weight Generation for Regional Wind-Induced Structural Response Estimation

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THE PROBLEM

Portfolio-scale wind resilience is computationally prohibitive.

0 s

OpenSees per building
single CPU core

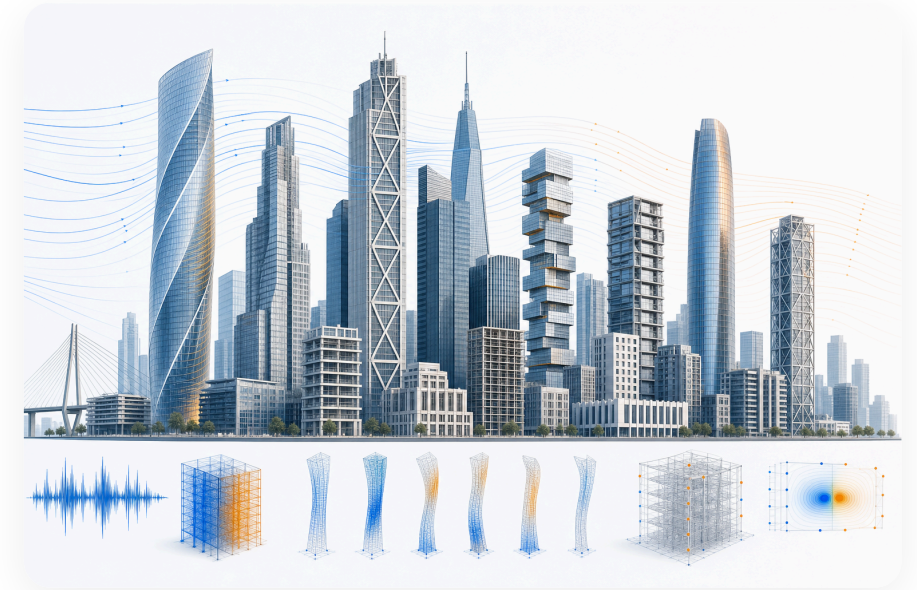
0

buildings in a
regional portfolio

0 h

wall-clock to simulate
900 test buildings

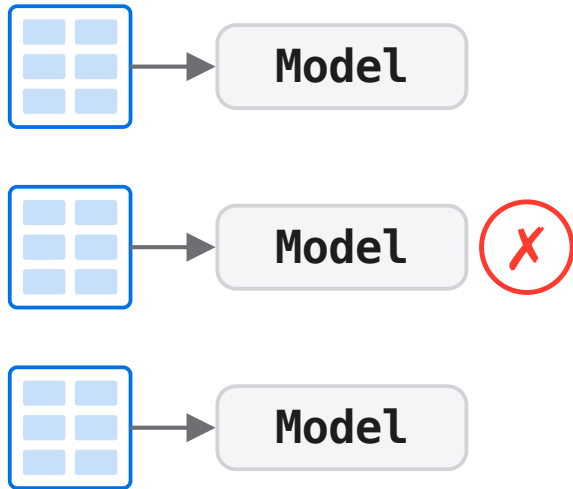
One nonlinear time-history analysis per building, per hazard scenario. Multiply by thousands of structures — direct simulation is impractical at portfolio scale.



Portfolio-scale wind assessment spans large, heterogeneous building inventories

Conventional surrogates don't transfer across buildings.

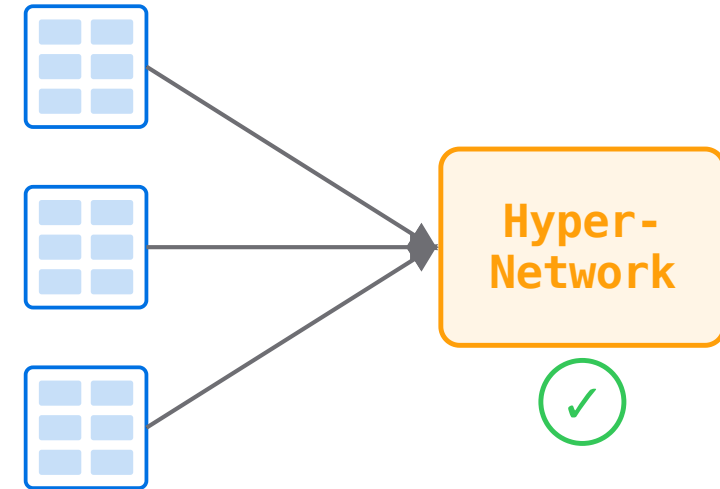
One model per building



N buildings → N trained models

VS

One shared HyperNetwork

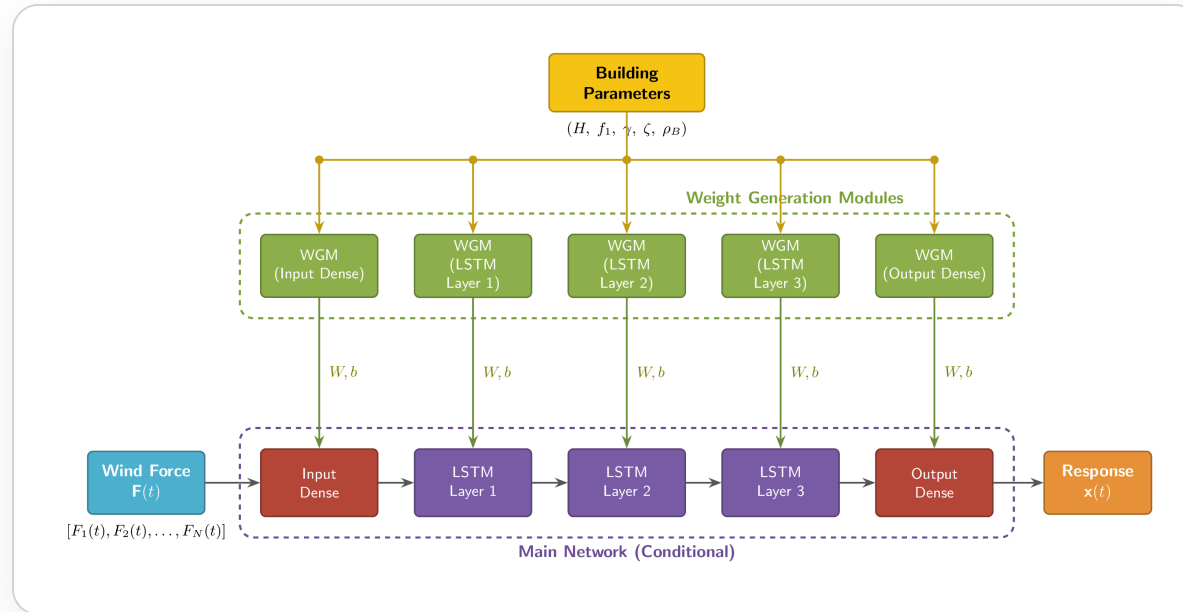


N buildings → 1 model, no retraining

Different **height, stiffness, damping, mass** —
one surrogate per building doesn't transfer

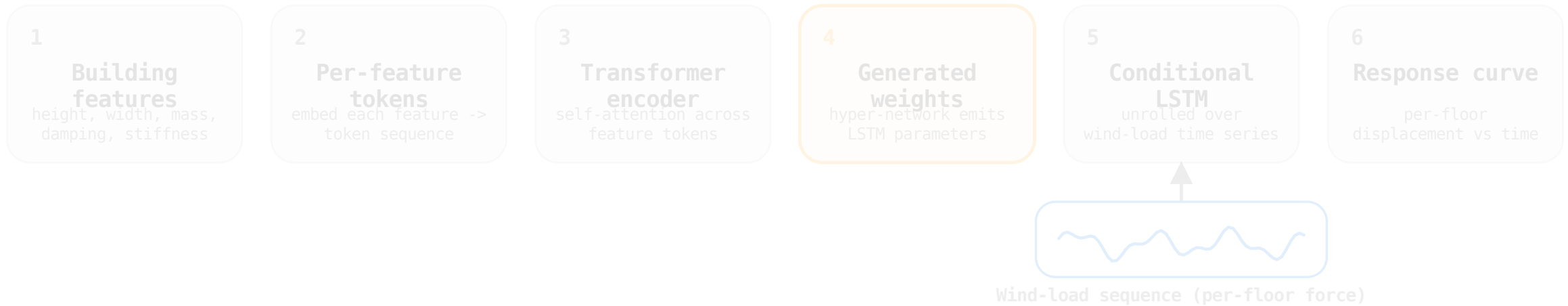
KEY IDEA

Generate the response network's weights from the building's feature vector.



One HyperNetwork reads building features and generates a dedicated response predictor — **one trained model, any building.**

From features to response — one forward pass.



Two inputs: building features generate the network; the wind-load sequence drives it to produce the response.

Transformer WGM

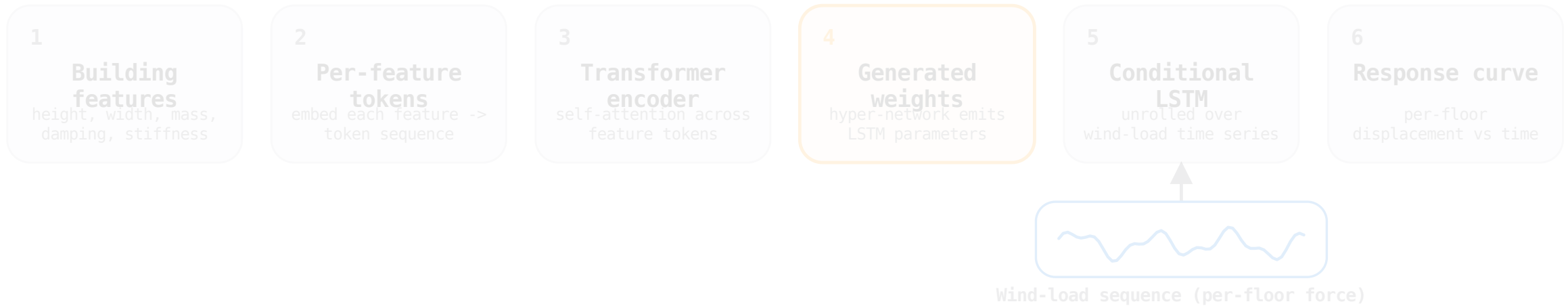
2 layers · dim 64 · 2 heads · FFN 256

Dense → LSTM×3 → Dense

5

Weights generated, not stored

From features to response — one forward pass.



Two inputs: building features generate the network; the wind-load sequence drives it to produce the response.

Transformer WGM

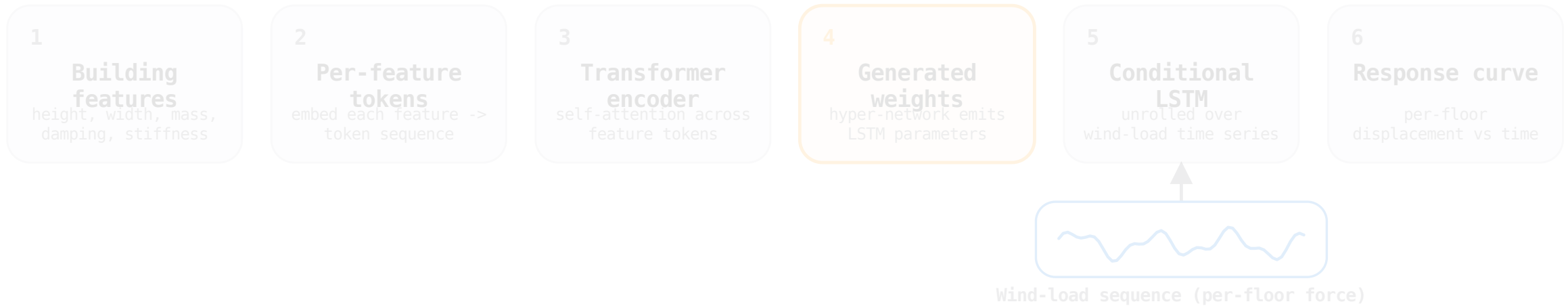
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5.1

Weights generated, not stored

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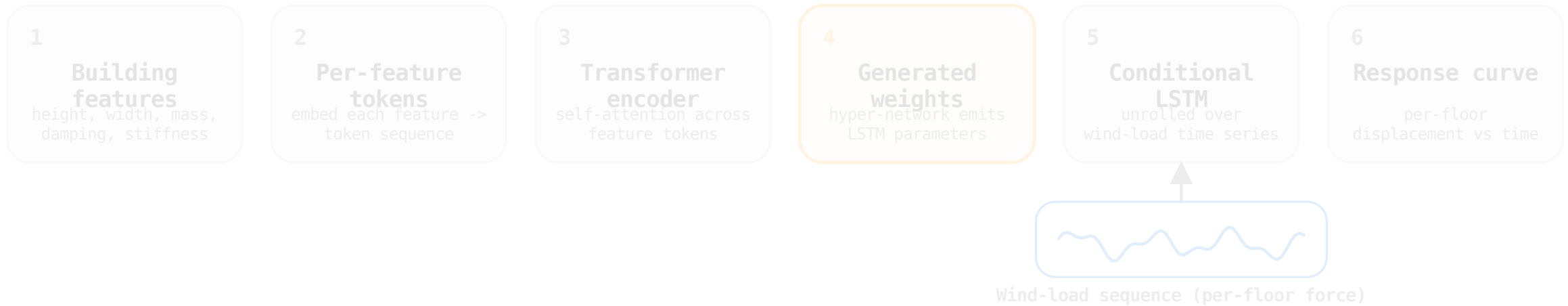
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Dense → LSTM×3 → Dense

5.2

Weights generated, not stored

From features to response — one forward pass.



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Transformer WGM

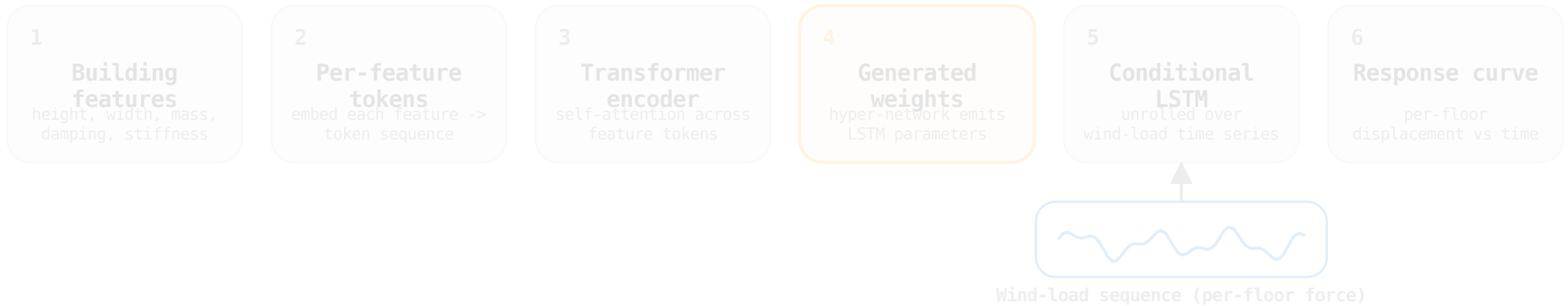
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Dense → LSTM×3 → Dense

5.3

Weights generated, not stored

From features to response — one forward pass.



Two inputs: building features generate the network; the wind-load sequence drives it to produce the response.

Transformer WGM

2 layers · dim 64 · 2 heads · FFN 256

Dense → LSTM×3 → Dense

5.4

Weights generated, not stored

From features to response — one forward pass.



Two inputs: building features generate the network; the wind-load sequence drives it to produce the response.

Transformer WGM

2 layers · dim 64 · 2 heads · FFN 256

Dense → LSTM×3 → Dense

5.5

Weights generated, not stored

A plain MLP generator collapses — the Transformer doesn't.

MLP WEIGHT GENERATOR

0.070

best RMSE (2 000 ep.)

Standard 500-ep. run collapses to flat / mean —
RMSE $\approx$ 0.116, output std < 5% of target

TRANSFORMER WEIGHT GENERATOR

0.012

RMSE — no collapse

Per-feature tokens + self-attention converge reliably,
every run, $\approx$ 500 epochs

MLP generator

0.070

Transformer generator

0.012

$\approx$ 6× better RMSE

No collapse

MLP needs 4× longer training to reach 0.070

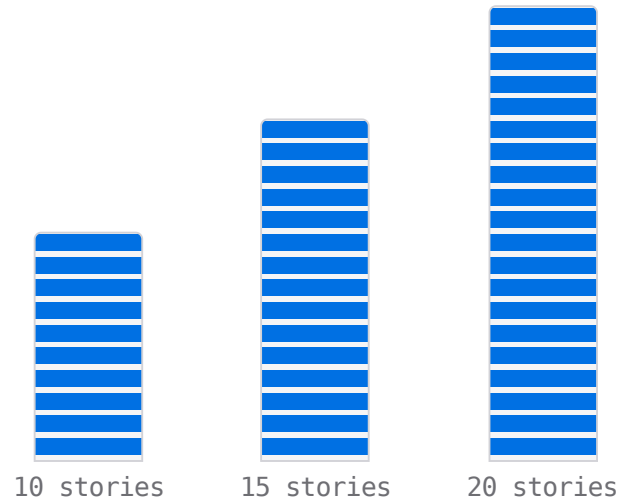
Per-feature tokens + self-attention capture cross-feature structure (e.g. how height interacts with damping). The MLP sees a flat feature vector and gets stuck predicting the mean — a loss-plateau failure mode.

Single-building diagnostic (RMSE on building 0) — separate from the 160-case benchmark NRMSE (0.017).

HEIGHT NORMALIZATION

10 / 15 / 20 stories — one shared 20-point grid.

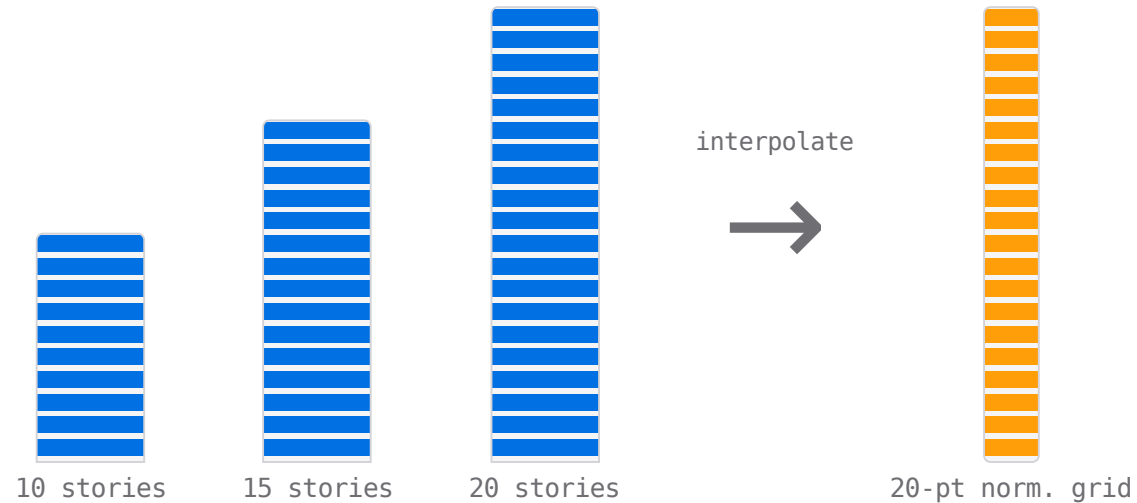
Map each building's floor responses to a fixed normalized-height axis so every output is always active.



HEIGHT NORMALIZATION

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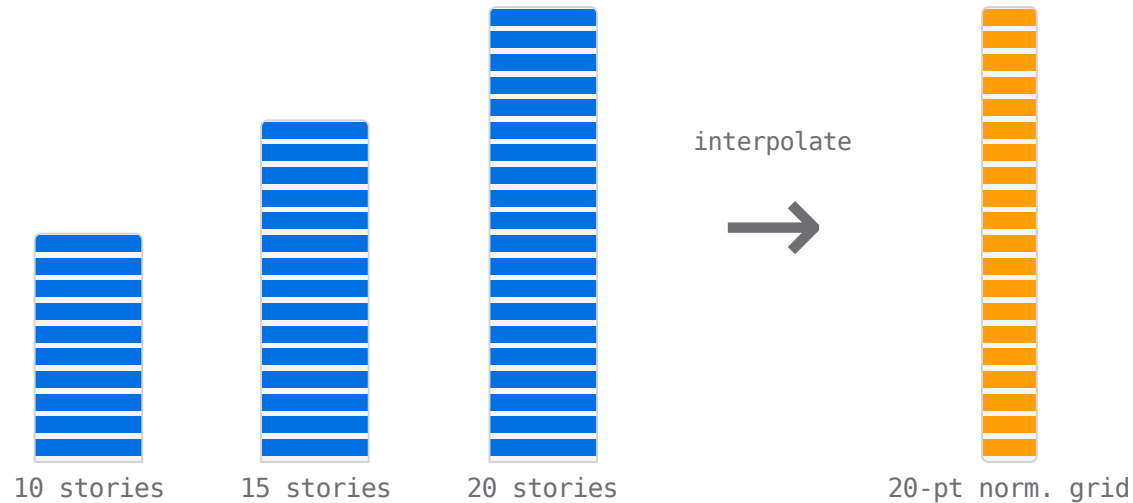
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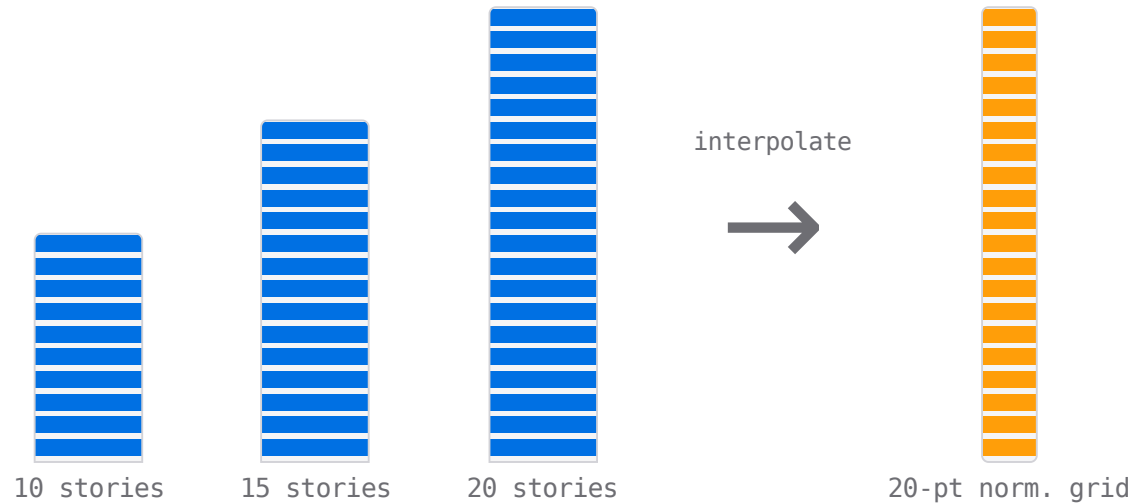
Zero-padding: inactive floors waste capacity

Interpolation: all 20 outputs always active

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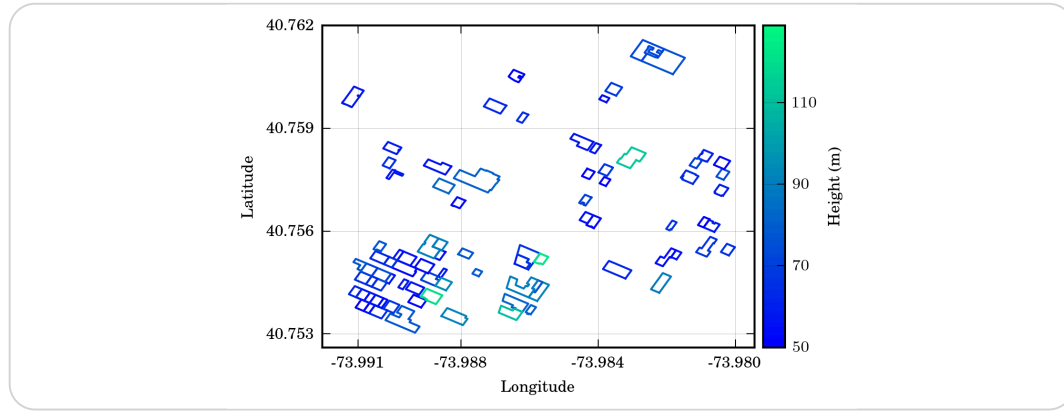
Zero-padding: inactive floors waste capacity

Interpolation: all 20 outputs always active

6.83 M parameters · population mean NRMSE: 1.05 % (10-story) · 1.01 % (15-story) · 0.88 % (20-story) — error is uniform across story counts

EXPERIMENTAL SETUP

Times Square-calibrated benchmark — scaled to 9,000.



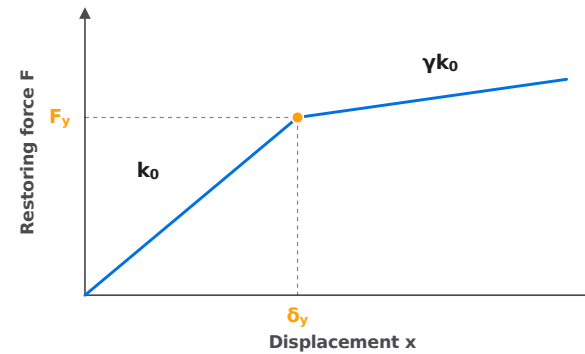
85 real buildings near Times Square (**OpenStreetMap**, 50-123 m) — the reference for the synthetic benchmark.

160 synthetic cases · 128/16/16 split

80 configs × 2 wind realizations

60 s wind records

BILINEAR STIFFNESS



F restoring force · x displacement

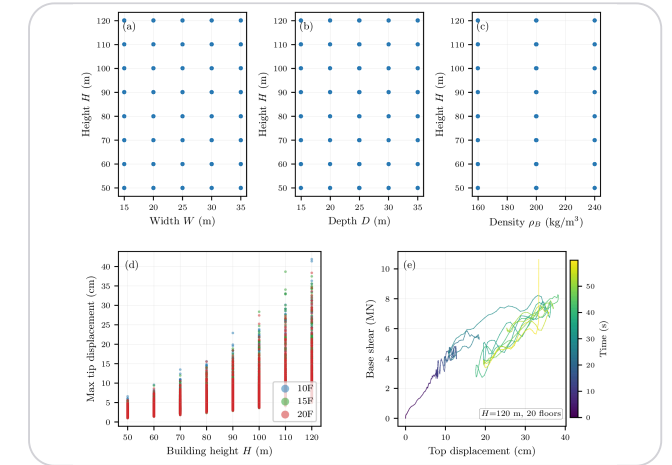
k_0 initial (elastic) stiffness

F_y yield force · δ_y yield displacement

γ post-yield ratio $\in [0.05, 0.10]$ · γk_0

post-yield stiffness

9,000-BUILDING FEATURE GRID



H height · W width · D depth

ζ damping ratio · ρ density

N_s number of stories · f fundamental frequency

Full-factorial over 7 features; training (blue) spans the test portfolio (red).

160-CASE TIMES SQUARE BENCHMARK

NRMSE 0.017 on 16 held-out test cases.

MEAN NRMSE

0.000

16 held-out test cases

160 synthetic cases — 128 train / 16 val / **16 test**
80 configs × 2 wind realizations. No test-case simulation in training.

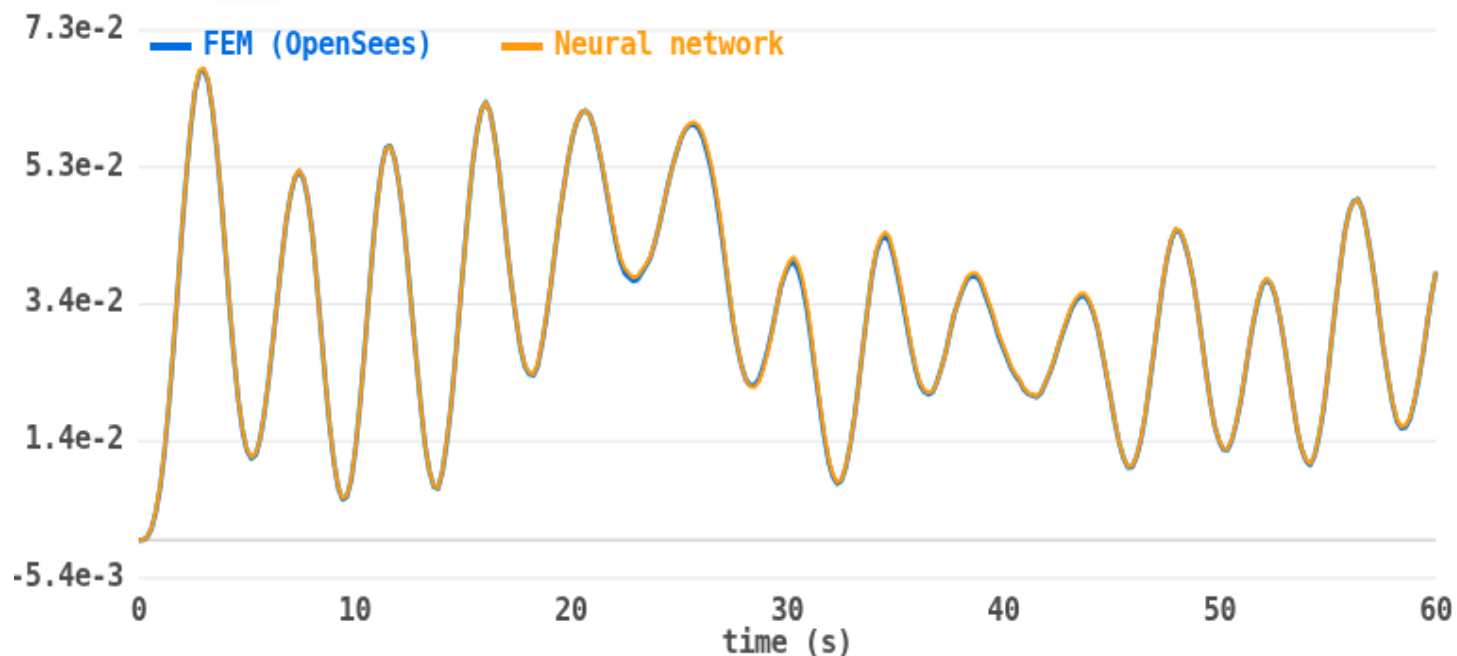
LOAD CASE

Wind

Constant

Sine

Play



FEM HyperNetwork

Demonstration building — H = 85 m, 15 floors, top-floor displacement

SCALE & GENERALIZATION – 9,000-BUILDING DATASET

Under 1% NRMSE on 900 held-out buildings.

0.00%

mean NRMSE
900 held-out buildings

0.00%

median NRMSE
900 held-out buildings

0

training buildings
building-level split

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UNIFORM ACROSS HEIGHTS

1.05%

10 floors

1.01%

15 floors

0.88%

20 floors

Height-normalized interpolation keeps error even across all building heights.

DURATION GENERALIZATION

60 s

training window



600 s

evaluated signal

Separate long-duration test — error stays low throughout the full 600-s signal.
Model learned structural dynamics, not a fixed template.

~7,000× faster than single-core OpenSees.

OpenSees FEM ~80 s / building

Neural network ~11.4 ms / building (batched)

~0x

faster per building

0 s

OpenSees / building
single CPU core



0x

HyperNetwork
batched GPU inference

≈ 20 h

OpenSees
portfolio of 900

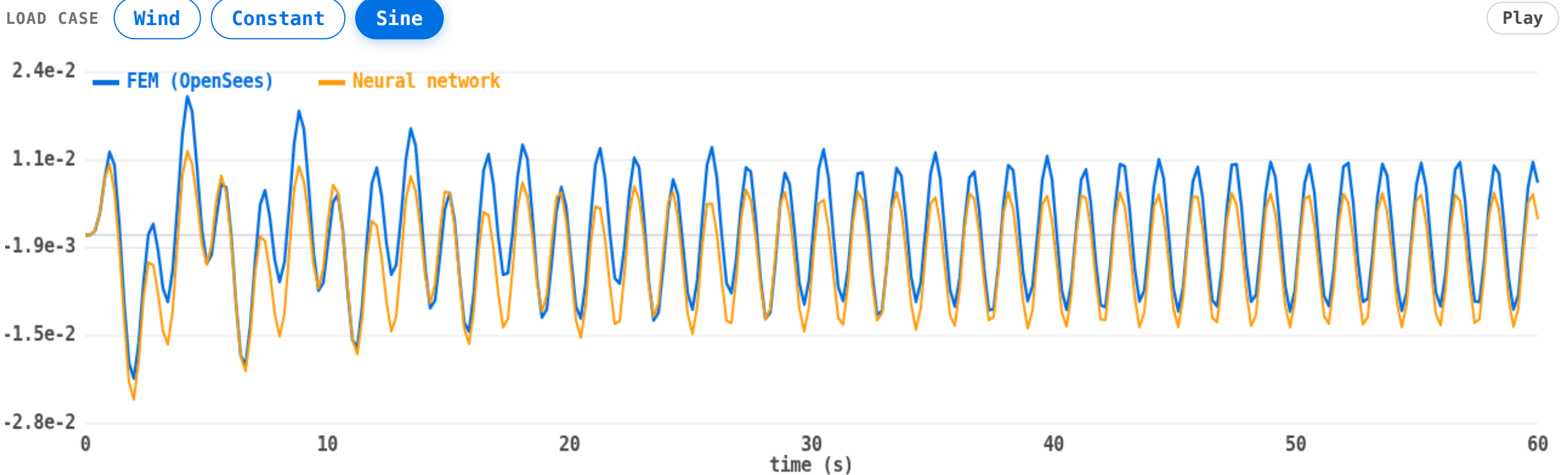


seconds

HyperNetwork
same 900 buildings

Probe any load — surrogate vs. FEM, on demand.

Probe any load class live — the surrogate verified against a fresh FEM run on demand. No cached data. Every verdict is a new execution.



Demonstration building (top-floor displacement) — H 85 m · 15F · ζ 0.02. Resonant sine shown. Switch load class with the controls above.

Multi-force input → multi-response output.

The surrogate is MIMO: forces at every floor map to displacements at every floor, matched against FEM.

LOAD CASE

Wind

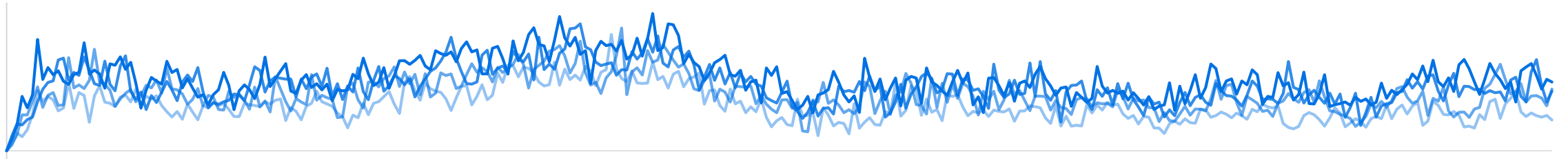
Constant

Sine

Play

Input — applied forces

per-floor — non-uniform

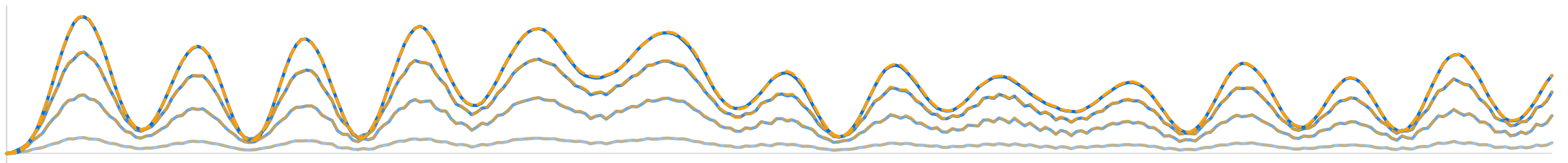


Output — floor displacements

FEM

model

floors 3 · 7 · 11 · 15 of 15



time (s)

BEYOND WIND

Targeted on wind — transferable to other loads.

Population mean NRMSE —
900 held-out buildings:

1.25 %

Wind

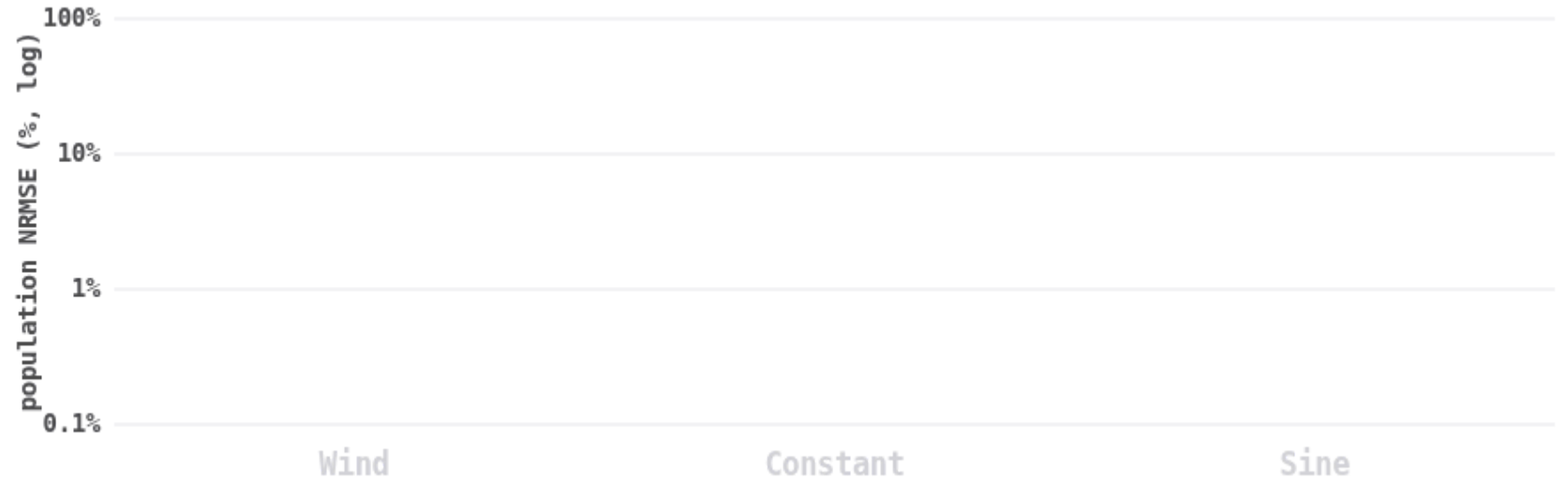
4.6 %

Constant

13.7 %

Sine (near-resonance)

Surrogate generalizes to unseen load types
without retraining.



Constant (4.6 %) and sine (13.7 %) confirm the surrogate generalizes beyond wind. Click to reveal each load class.

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Future directions.

- **More complex loads — the immediate next step.** Extend to blast, multi-hazard, and arbitrary broadband excitations.
- **Physics-informed / knowledge-enhanced training** — embed governing equations to improve extrapolation and cut data demand for larger, more diverse inventories.
- **Richer features + 3D models** — image and BIM-driven building representations; extend to 3D FEM with torsional and complex nonlinear behavior.

One model, many buildings.

- **Cross-building generalization** — one trained model serves a heterogeneous portfolio with no per-building retraining.
- **~7,000× faster** than OpenSees (~80 s/building, single CPU core).
- **<1 % median NRMSE** on 900 held-out buildings (0.89 %).
- **Portfolio resilience + digital twins** — regional screening and scenario analysis now tractable on ordinary GPU hardware.

7,000×
speedup over OpenSees

0.89 %
median NRMSE · 900 test buildings

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Thank you.

Questions?

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